Geometry-Aware, Language-Conditioned Predictive Safety for Mobile Robot Navigation

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Abstract

Problem: Dynamic, context-dependent safety (often language-specified in human-centered contexts)

Gap: Fixed safe sets; semantically blind (uniform obstacle treatment)

Framework:

- Convert natural language into structured, perception-grounded constraints
- Semantic safety fields encode spatial, context-dependent behavior
- MPC (planning) + CBF (real-time guarantees)

Outcome: Interpretable, context-aware behavior with real-time tractability

Introduction

Safe set is defined as the superlevel set of a function $h(x)$:

$$S = \{x \in \mathbb{R}^n \mid h(x) \geq 0\}$$

A control-affine system:

$$\dot{x} = f(x) + g(x)u,$$

Control Barrier Function enforces **forward invariance**:

$$\nabla h(x)^\top f(x) + \nabla h(x)^\top g(x)u + \alpha\big(h(x)\big) \geq 0$$

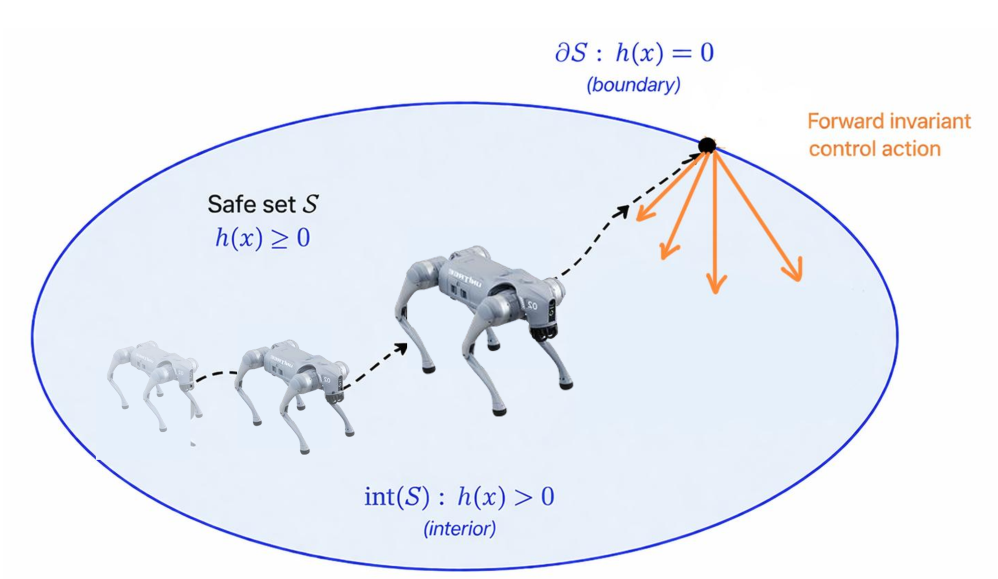

Fig. 1. Forward Invariance Definition

We synthesize $h(x)$ using a **Poisson Safety Function**:

$$\Delta h(x) = f_p(x), h(x) = 0 \text{ on } \partial\Omega$$

This yields a smooth, geometry-aware safety field that can be enforced with MPC+CBFs.

Initial Results

Semantics-Aware Safety Algorithm:

- Language-conditioned safety fields enable **context-aware behavior** with real-time, **formal guarantees**.
- Semantic perception mapped to safety fields to guide and constrain motion.

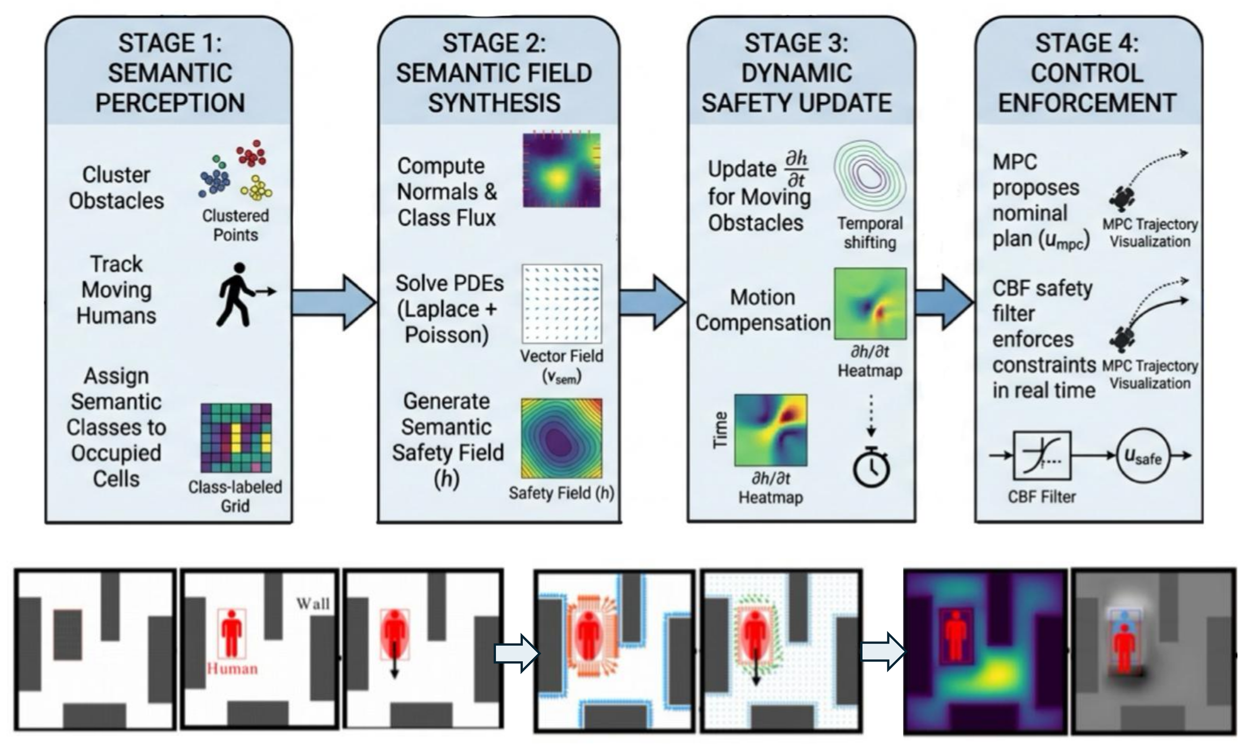

Fig. 2. Semantic Safety Pipeline and Block Diagram

Context-Aware Safety Fields:

Spatially-varying safety fields conditioned on semantics:

- Human obstacles induce larger repulsive regions
- Static objects allow tighter, task-efficient navigation
- **Safety representation adapts** across the environment

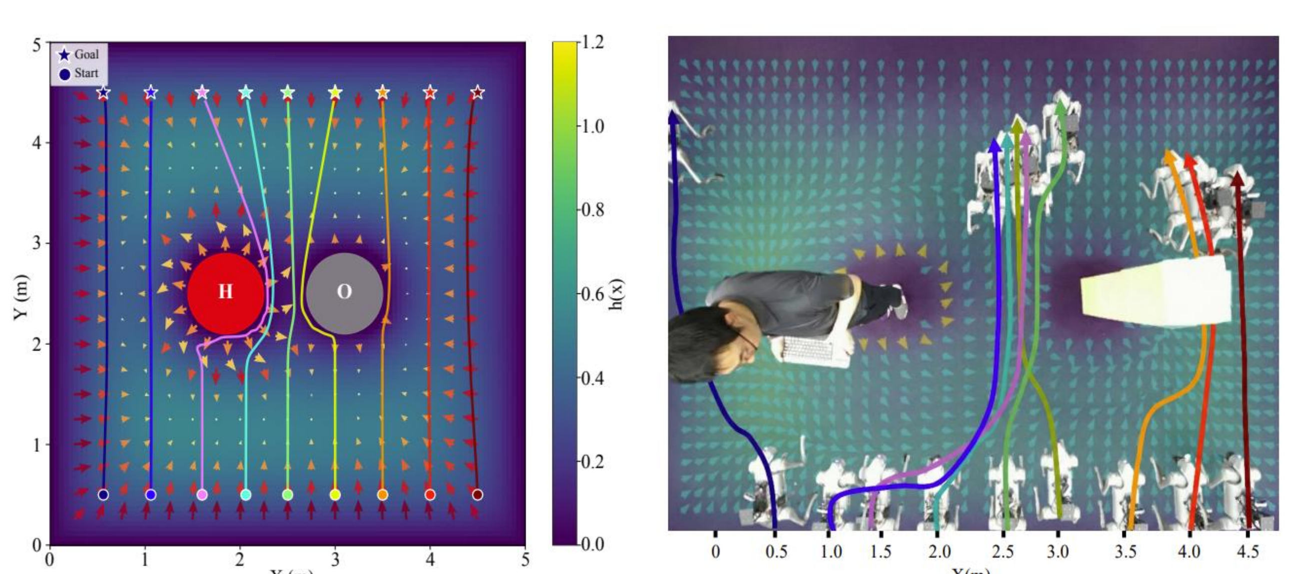

Fig. 3. Simulation and hardware experiment exhibiting social compliance and context-aware safety fields.

Safety evaluated in constrained human-interaction scenarios using **human–robot margin**, and **max lateral offset**.

Metric	Proposed	Baseline
Human-Robot (m)	0.318 ± 0.0774	-0.008 ± 0.0625
Max Offset (m)	0.75	-0.1

Dual-Layer Safety:

Model predictive control (MPC) and control barrier functions (CBFs) are combined to **enforce safety**.

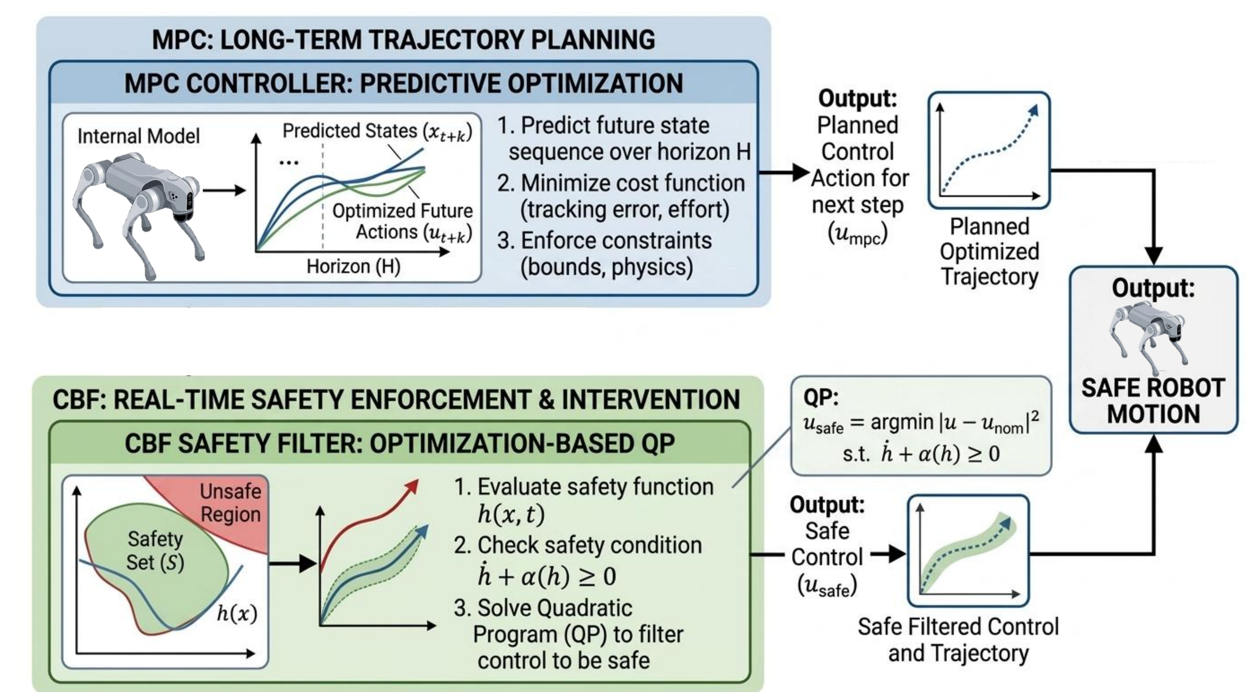

Fig. 4. MPC+CBF Safety Layers Implementation

Simulation and Hardware Deployment:

Validated in simulation and on Unitree Go2 and G1 across hallways, cluttered human environments, and open spaces **(Fig. 5–6).**

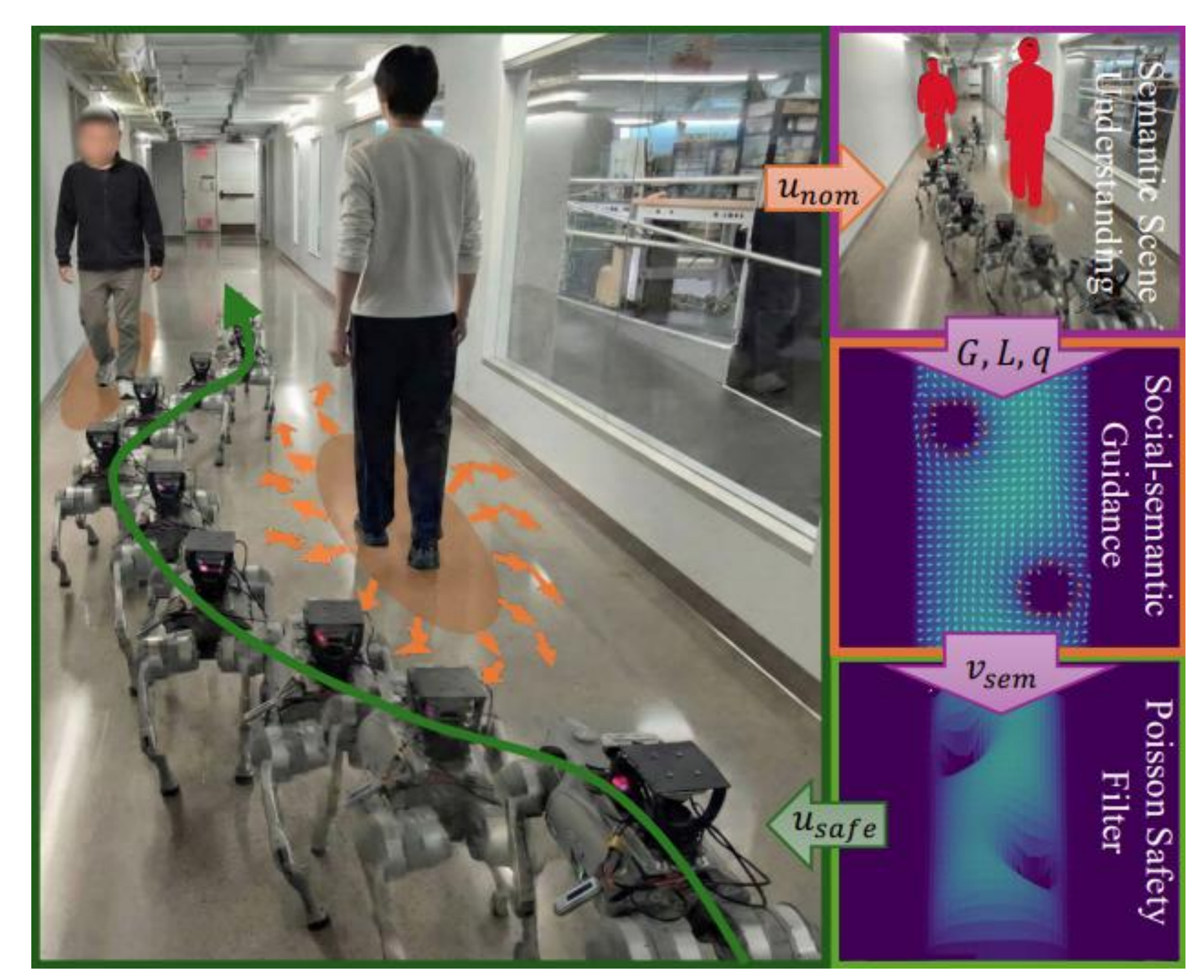

Fig. 5. Hardware experiment: UnitreeGo2 maintains buffer from humans while navigating obstacles.

Poisson Safety Fields (PSFs):

HJ reachability requires high-dimensional value functions, enables strong safety guarantees but **computationally expensive and hard to adapt.**

Poisson safety field: more efficient and tractable, enabling **semantic and geometric context** under real-time constraints.

Conclusion

We present a framework that generates geometry-aware, semantics-conditioned safety fields and enforces them with MPC + CBF, enabling **context-aware behavior with formal safety guarantees.**

Key contributions:

- **Semantic safety representation**
- **Experimental Validation**
- **Real-time deployment**

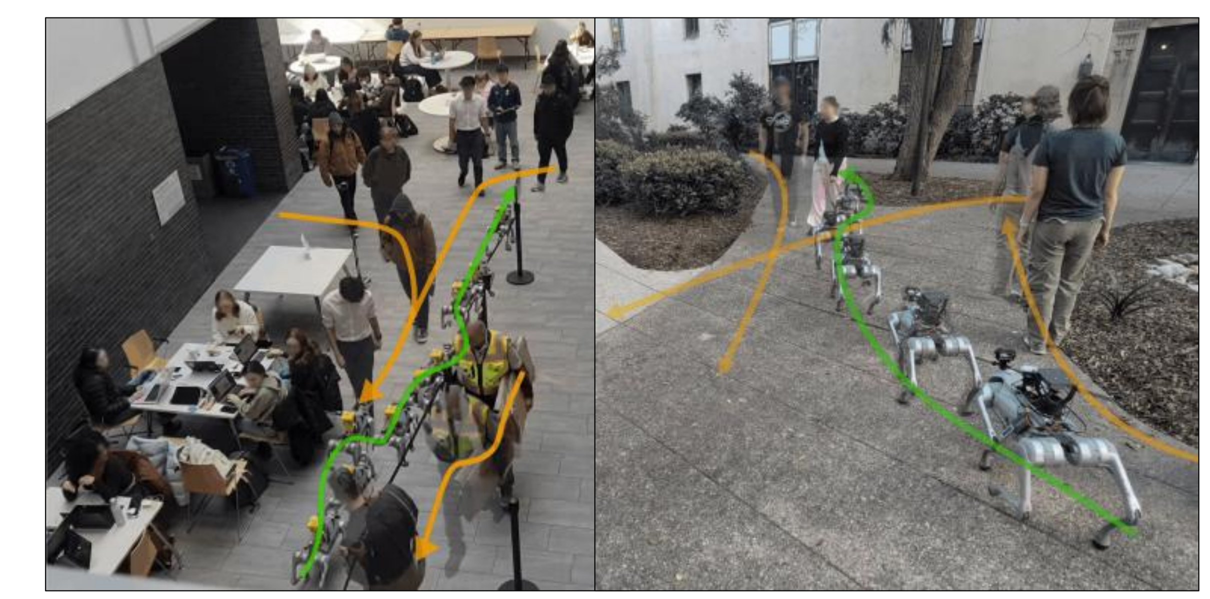

Fig. 6. Safe-SAGE algorithm onboard Unitree Go-2

This work lays the foundation for **language-conditioned safety,** where high-level instructions are translated into structured, enforceable constraints.

Future Work

Future work focuses on extending the framework to **language-conditioned safety** in dynamic environments:

- **Language Interface:** Structured, interpretable constraints (objects, types, margins)
- **Performance Optimization:** Further reduce latency via continued profiling
- **Richer Semantic Integration:** Enhance robustness with richer, generalizable semantic representations

Paper Pre-print

Video Demo